



12 UCI Datasets — 6 Medical, 6 Agricultural ($n = 900\text{--}14,980$)



CASH Search Space — Scaler $\times$ Imbalance Handling $\times$ Feature Selection $\times$ 9 Model Families

f_1 : Predictive Performance
— MCC



$f_2 = \Phi$: Explanation Faithfulness
— SHAP Insertion/Deletion

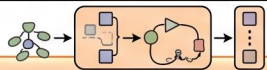


f_3 : Inference Cost
— Structural Proxy $c(\mathbf{x})$



Search Method

f_1, f_2, f_3

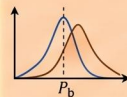


NSGA-II (MOXEC) — 3 objectives, 100 trials $\times$ 3 seeds

Random Search (3-objective)

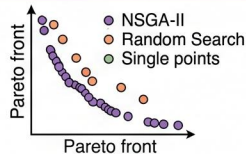
Search Method

f_1 only

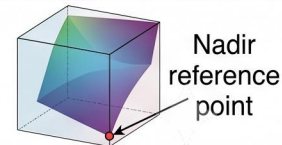


TPE, FLAML, AutoGluon — MCC only

Pareto Front P^* (NSGA-II, Random Search) or Degenerate Single-Point Front (TPE, FLAML, AutoGluon)



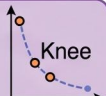
Hypervolume $HV(P^*, r)$ vs. Union-Based Nadir Reference Point



Nadir
reference
point



Statistical Testing — Friedman/Nemenyi,
Wilcoxon, Mann-Whitney



Front Navigation — Knee-Point,
Latency-Constrained Best-MCC, Weighted MES

